1 Solutions of Problems in Exercise Session 6

Exercise 1. Explain why reinforcement learning is not used for full training of LLMs. What would the underlying MDP look like? How does the fine-tuning stage differ?

Solution. To train a model using RL, rewards are required, which are not readily available during the initial training stage of LLMs. Supervised learning is more robust and efficient for this phase, enabling the training of massive models on vast datasets. While various MDP formulations could be considered, the one relevant to RLHF involves states as prompts sampled in an independent and identically distributed (i.i.d.) manner (where the next state is independent of the current one), actions as possible responses, and rewards based on human preferences for these responses. The fine-tuning stage differs in that the dataset is much smaller, rewards can be learned from preference data, and the process begins with a strong initial policy or model.

Exercise 2. You want to learn a preference model for Swedish pastries. You conduct a survey asking people whether they prefer *kannelbulle* (ℓ) over *kardemummabulle* (m), and you collect 25 responses of the form (ℓ, m) (indicating preference for *kannelbulle*) and 10 of the form (m, ℓ). What probability model would minimizing the loss above yield?

Solution. Recall that the optimal probability model for dataset $\mathcal{D}$ is found by minimizing the loss:

$$\min_P \mathcal{L}_P(\mathcal{D}) = \min_P \Big[-\sum_{i=1}^{|\mathcal{D}|} \log P(y_w^i \succ y_\ell^i) \Big]$$

Here, $|\mathcal{D}| = 35$, with $\ell^i \succ m^i$ for $i = 1, \ldots, 25$, and $m^i \succ \ell^i$ for $i = 26, \ldots, 35$. The goal is to maximize:

$$\max_P \sum_{i=1}^{25} \log P(\ell \succ m) + \sum_{i=26}^{35} \log \underbrace{P(\ell \succ m)}_{1-P(m\succ\ell)}$$

where we assumed that either $\ell \succ m$ or $m \succ \ell$ (you have to have a preference for one of the two) and thus $P(\ell \succ m) + P(m \succ \ell) = 1$. The last equation can be written as $25 \log P(\ell \succ m) + 10 \log(1 - P(\ell \succ m))$, and optimizing over $P(\ell \succ m)$ gives $P(\ell \succ m) = \frac{25}{35}$, as expected, since 25 out of 35 individuals preferred ℓ over m.

Exercise 3. Repeat Exercise 2, but this time suggest rewards $r(m)$ and $r(\ell)$ that model preferences of surveyed individuals according to the Bradley-Terry model. Are the rewards uniquely defined?

Solution. Using the probability $P(\ell \succ m)$ derived in Exercise 2, we can immediately compute rewards $r(m)$ and $r(\ell)$. But for the sake of practice, let's ignore Exercise 2, and derive the rewards directly. We solve:

$$\max_r \mathbb{E}_{(y_w, y_\ell)\sim\mathcal{D}} \Big[\log \sigma\Big(r(y_w) - r(y_\ell) \Big) \Big] = \max_r \sum_{i=1}^{25} \log \sigma(r(\ell) - r(m)) + \sum_{i=26}^{35} \log \sigma(r(m) - r(\ell))$$

where $y_w, y_\ell \in \{\ell, m\}$. Denoting $\exp(r(m) - r(\ell)) = x$, the objective becomes:

$$\max_r \sum_{i=1}^{25} \log \sigma(r(\ell) - r(m)) + \sum_{i=26}^{35} \log \sigma(r(m) - r(\ell))$$
$$= \max_x 25 \log \frac{1}{1+x} + 10 \log \frac{1}{1+\frac{1}{x}}$$

Solving yields $x = \frac{2}{5}$, giving $r(\ell) = r(m) + \log \frac{5}{2}$. The rewards are uniquely defined up to an additive constant.

Exercise 4. Human feedback for every time step and each action may be difficult to collect. Instead, human evaluators provide preferences for trajectories τ of a fixed length T, ranking them as $\tau^i \succ \tau^j$ if trajectory τ^i is preferred over τ^j. How would you use the Bradley-Terry model to learn rewards from such trajectory-based preferences?

Solution. In this case, we can use the following model, which considers all the states and actions encountered in the trajectories:

$$P(\tau^i \succ \tau^j) = \frac{\exp\left(\sum_{(s_t, a_t) \in \tau^i} r(s_t, a_t)\right)}{\left(\sum_{(s_t, a_t) \in \tau^j} r(s_t, a_t)\right) + \left(\sum_{(s_t, a_t) \in \tau_\ell} r(s_t, a_t)\right)}$$

This formulation ensures that the probabilities are normalized and interpretable as a comparison of cumulative rewards across trajectories.

Exercise 5. Propose a reward proxy $r(s, a)$ to replace the reward using the objective above. Suggest how PPO could be used to optimize the policy π_θ. Explain the difference between π_{ref} and π_{old} in the PPO objective. Is it problematic that PPO is an on-policy algorithm?

Solution. Recall that the objective is given by:

$$\max_\theta \mathcal{L}_\pi(\theta, \mathcal{D}) = \max_\theta \mathbb{E}_{s \sim \mathcal{D}} \left[\mathbb{E}_{a \sim \pi_\theta(\cdot|s)}[r_\phi(s, a)] - \beta D_{\text{KL}}(\pi_\theta(\cdot|s) || \pi_{\text{ref}}(\cdot|s))\right]$$
$$= \max_\theta \mathbb{E}_{s \sim \mathcal{D}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[r_\phi(s, a) - \beta \log \frac{\pi_\theta(a|s)}{\pi_{\text{ref}}(a|s)}\right]$$

However, when deriving the PPO algorithm, we considered optimizing an objective like $\mathbb{E}_{s \sim \mathcal{D}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} Q^{\pi_\theta}(s, a)$. In cases where the next state is independent of the previous state (i.e. $P(s_{t+1}|s_t) = P(s_{t+1})$), this objectives reduces to $\mathbb{E}_{s \sim \mathcal{D}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} r(s, a)$. We can assume that preference data has been collected in an i.i.d. manner (ex. you give preference feedback to ChatGPT sporadically, every few days, rather than sequentially, after each prompt). Since we know how to optimize policies when the objective is $\mathbb{E}_{s \sim \mathcal{D}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} r(s, a)$, we can define a new reward proxy $\overline{r}(s, a)$ as:

$$\overline{r}(s, a) = r_\phi(s, a) - \beta \log \frac{\pi_\theta(a|s)}{\pi_{\text{ref}}(a|s)},$$

so that the objective can equivalently be written as: $\mathbb{E}_{s \sim \mathcal{D}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \overline{r}(s, a)$, a standard form suitable for applying RL algorithms like PPO.
Note that π_{ref} is a fixed, pretrained reference policy that we aim to align with. It remains unchanged during training. On other hand, π_{old} in PPO algorithm changes iteratively, as

each new policy becomes the "old" policy for the next update. Both constraints ensure the policy does not deviate too far, but they serve difference purposes. π_{ref} aligns the policy with a fixed target, while π_{old} ensures stable updates during training.
While PPO is an on-policy algorithm, this does not pose a significant limitation in this context. Since we can compute the reward for any state-action pair (s, a) using the learned reward model $r_\phi(s, a)$, we are not constrained by the policy used to generate the data. Furthermore, because states are treated as independent, the choice of policy used when this state was encountered is not important.

Exercise 6. The Figure on the left reproduces Figure 5 from [7]. Explain what this figure illustrates. This phenomenon is known as reward hacking - what do you think why? Does the figure suggest that fine-tuning improves performance compared to the reference model? How do you think the summaries generated by this overtrained LLM might look? (you can find them in Table 29 in [7]!)

Solution. The figure illustrates that when the model deviates significantly from the reference model (indicated by a large KL divergence from the supervised baseline), the reward model becomes less effective. This occurs because the learned reward model is imperfect: some state-action pairs are unobserved, and the reward model might assign arbitrary (potentially very high) rewards to these pairs. Consequently, optimizing a policy to maximize these rewards may result in learning a policy that exploits these unrealistically high rewards.
However, the figure also shows that for smaller KL divergences (up to approximately 10 in this setting), the policy indeed improves upon the reference (baseline).

Exercise 7. Fix a state s. Consider optimizing $\max_\theta \mathcal{L}^s_\pi(\theta, \mathcal{D})$ under the constraint that $\sum_a \pi_\theta(a|s) = 1$. Solve this optimization problem using Lagrange multipliers and show that the optimal solution is:

$$\pi^\star_\theta(a|s) = \frac{1}{Z(\beta, s)} \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} r_\phi(s, a)\right)$$

where $Z(\beta, s)$ is the partition function given by: $Z(\beta, s) := \sum_a \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} r_\phi(s, a)\right)$.

Solution. We aim to maximize the following objective:

$$\mathcal{L}^s_\pi(\theta, \mathcal{D}) = \mathbb{E}_{s\sim\mathcal{D}}\left[\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[r_\phi(s, a)] - \beta D_{\text{KL}}(\pi_\theta(\cdot|s)||\pi_{\text{ref}}(\cdot|s))\right]$$

subject to the constraint $\sum_a \pi_\theta(a|s) = 1$. We define the corresponding Lagrangian as:

$$\begin{aligned}\mathcal{L}(\pi_\theta, r_\phi) &= \mathcal{L}^s_\pi(\theta, \mathcal{D}) + \lambda\left(\sum_a \pi_\theta(a|s) - 1\right)\\ &= \sum_a \pi_\theta(a|s)\Big(r_\phi(s, a) - \beta \log \pi_\theta(a|s) + \beta \log \pi_{\text{ref}}(a|s)\Big) + \lambda\left(\sum_a \pi_\theta(a|s) - 1\right)\end{aligned}$$

Taking the derivative with respect to $\pi_\theta(a|s)$ and setting it to zero:

$$r_\phi(s, a) - \beta \log \pi_\theta(a|s) - \beta + \beta \log \pi_{\text{ref}}(a|s) + \lambda = 0$$

Rearranging this we obtain:

$$\pi_\theta(a|s) = \pi_{\text{ref}}(a|s) \exp\left(\frac{\lambda}{\beta} - 1\right) \exp\left(\frac{1}{\beta} r_\phi(s, a)\right)$$

Using the constraint $\sum_a \pi(a|s) = 1$, we find:

$$\exp\left(1 - \frac{\lambda}{\beta}\right) = \sum_a \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} r_\phi(s,a)\right) = Z(\beta, s)$$

and thus $\lambda = \beta(1 + \log Z(\beta, s))$, where $Z(\beta, s)$ is the partition function. Substituting λ back into equation for $\pi_\theta(a|s)$ yields:

$$\pi_\theta(a|s) = \frac{1}{Z(\beta, s)} \pi_{\text{ref}}(a|s) \exp\left(\frac{1}{\beta} r_\phi(s,a)\right)$$

as required.

Exercise 8. Consider the setting of Exercise 2. This time, suppose that $P(\ell \succ m) = 0.9$ and $P(m \succ \ell) = 0.1$. On a cold December morning, you manage to survey only two people, and both prefer ℓ. What would the optimal DPO-policy be in this case? Does it depend on the regularization parameter β? How would this change if using RLHF?

Solution. In this example, since both individuals prefer ℓ to m, the DPO-loss simplifies to:

$$\max_\pi 2 \log \sigma\Big(\beta \log \frac{\pi(\ell)}{\pi_{\text{ref}}(\ell)} - \beta \log \frac{\pi(m)}{\pi_{\text{ref}}(m)}\Big),$$

Nothing prevents us from setting $\pi(m) = 0$ and $\pi(\ell) = 1$, regardless of the reference policy π_{ref} or the regularization parameter β. This happens, even though $P(m \succ \ell) = 0.1 > 0$.
In contrast, using RLHF introduces regularization through the reward model. Instead of learning the exact optimal rewards (as in DPO), RLHF tends to learn rewards that are good but potentially suboptimal, particularly in more complex environments. This ensures that the learned policy does not set $\pi(m) = 0$.
In a nutshell, using optimal rewards $r^\star_\phi$ in DPO might make the algorithm less stable if not addressed properly (ex. early stopping DPO can help with this).